

## A146 · Comparative Analysis of ZRS–SHH Genomic Architecture Across Vertebrates

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Distal enhancers can regulate developmental genes over large genomic distances, but the extent to which enhancer–gene genomic architecture is conserved during vertebrate evolution remains unclear. The Zone of Polarizing Activity Regulatory Sequence (ZRS) is a well-characterized limb enhancer that regulates Sonic Hedgehog (SHH) expression, providing a model for investigating the evolution of long-range gene regulation. Here, we developed a comparative-genomics framework to map the human ZRS enhancer and the SHH promoter across Vertebrate Genomes Project (VGP) assemblies and quantify their genomic distance across species. We evaluated 577 VGP species and classified cross-species mappings according to whether homologous regions mapped to a single chromosome, multiple alignment blocks, or multiple chromosomes/scaffolds. We retained species in which both ZRS and the SHH promoter mapped to the same chromosome for downstream distance analysis. Among these species, we observed substantial variation in ZRS–SHH genomic distance across vertebrate lineages, with distinct distance distributions among mammals, birds, reptiles, and amphibians. We further examine the relationship of the ZRS–SHH distance and evolutionary divergence from humans and assess the distance relative to chromosome-scale genomic organization. This analysis provides a comparative framework for distinguishing conserved from divergent aspects of long-range enhancer–gene genomic architecture. By integrating cross-species sequence mapping with evolutionary and genomic-context information, we aim to determine whether the long-range genomic relationship between ZRS and SHH is evolutionarily constrained and to identify patterns of regulatory architecture associated with the conservation of long-range gene regulation.